

# MADHAV MERUVA

AI/ML Engineer | LLM & Backend Systems

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## SUMMARY

Computer Science undergraduate specializing in **Applied AI, LLM systems, and backend engineering**, with hands-on experience building **semantic search** platforms, **retrieval-augmented** systems, and **end-to-end ML** pipelines. Experienced in developing production-ready AI applications using **LangChain, Vector Databases, PyTorch, TensorFlow, and FastAPI**. Built scalable AI-powered systems with **real-time inference, caching optimization, and modular API architectures**. Seeking Applied AI / ML Engineering internship opportunities.

## EDUCATION

**Keshav Memorial Institute of Technology (KMIT)**

Bachelor of Technology, Computer Science Engineering | 2023 - 2027 | CGPA - 8.5

## PROJECTS

**AI Startup Idea Validator (LLM-Powered Evaluation Platform)** | **GitHub:** github.com/mad674/startupideavalidator

- Built an LLM-powered startup evaluation platform integrating **semantic retrieval, vector search**, and AI-generated business insights.
- Implemented retrieval pipelines using **LangChain and PineconeDB** to improve contextual reasoning and relevance of generated responses.
- Improved inference performance by ~30% using **Redis caching, optimized API** orchestration, and scalable backend workflows.
- Designed **modular AI service architecture** supporting real-time interactions, retrieval-augmented generation, and concurrent user requests.

**Technologies:** React.js, Node.js (Express), FastAPI, MongoDB, PineconeDB, Redis, LangChain, OpenAI API, JWT

**Financial Question Answering System (FinQA)** | **GitHub:** github.com/mad674/qa

- Built a **neuro-symbolic financial question-answering system** using a BERT retriever and LSTM-based program generator, and symbolic execution engine.
- Engineered **complete ML workflows** including preprocessing, model training, evaluation, and inference.
- Achieved **~80% accuracy** with interpretable program-step reasoning and domain-specific symbolic execution.

**Technologies:** Flutter, FastAPI, PyTorch, Firebase

**Jewellery Design Pattern Generator (Pix2Pix GAN)** | **GitHub:** github.com/mad674/mlmodel

- Built a **Pix2Pix GAN-based system for sketch-to-image generation** with real-time inference capabilities.
- Developed a **real-time pipeline for generating design previews** from user-uploaded sketches.
- Improved output quality through **dataset preprocessing** and training optimization techniques.

**Technologies:** React.js, Node.js (Express), JWT, MongoDB, TensorFlow, Flask

## SKILLS

- Languages:** Python, JavaScript, Java, C++
- AI/ML:** PyTorch, TensorFlow, NLP, Transformers, LangChain, Vector Databases
- Backend:** Node.js (Express), FastAPI, Flask, REST APIs
- Databases:** MongoDB, PineconeDB, Redis
- Cloud & DevOps:** Git, Postman, AWS, Docker
- Core Concepts:** Machine Learning, Deep Learning, DSA, OOP

## ACHIEVEMENTS AND CERTIFICATES

- Solved **500+ DSA** problems with consistent practice in problem solving, algorithms, and data structures.
- Secured a **top team award** at the KMIT Project Expo